

blehound1— 3x nRF54LM20 synchronized multi-channel sniffer (real parts; SoC/RF/footprints from the official DK)

Node A (ch37)

文件: node_a.kicad_sch

Node B (ch38)

文件: node_b.kicad_sch

Node C (ch39)

文件: node_c.kicad_sch

4 sub-sheets: Node A/B/C (each has SoC + RF + on-board antenna, guarding advertising channels 37/38/39) + Baseboard (USB/power). Double-click a block to enter.

Baseboard (USB hub / USB-C / Power / SYNC)

文件: board.kicad_sch

Inter-board SPI: independent point-to-point pairs, no shared bus. 3 edges of the triangle: A master-B slave / B master-C slave / C master-A slave;

Each board has 2 SPI ports (one master, one slave); each link adds a REQ line so the slave end can actively ask the master to read;

Each link always has exactly one master => zero arbitration. SYNC open-drain + 4.7k + shared GND; strap hard-codes the channel.

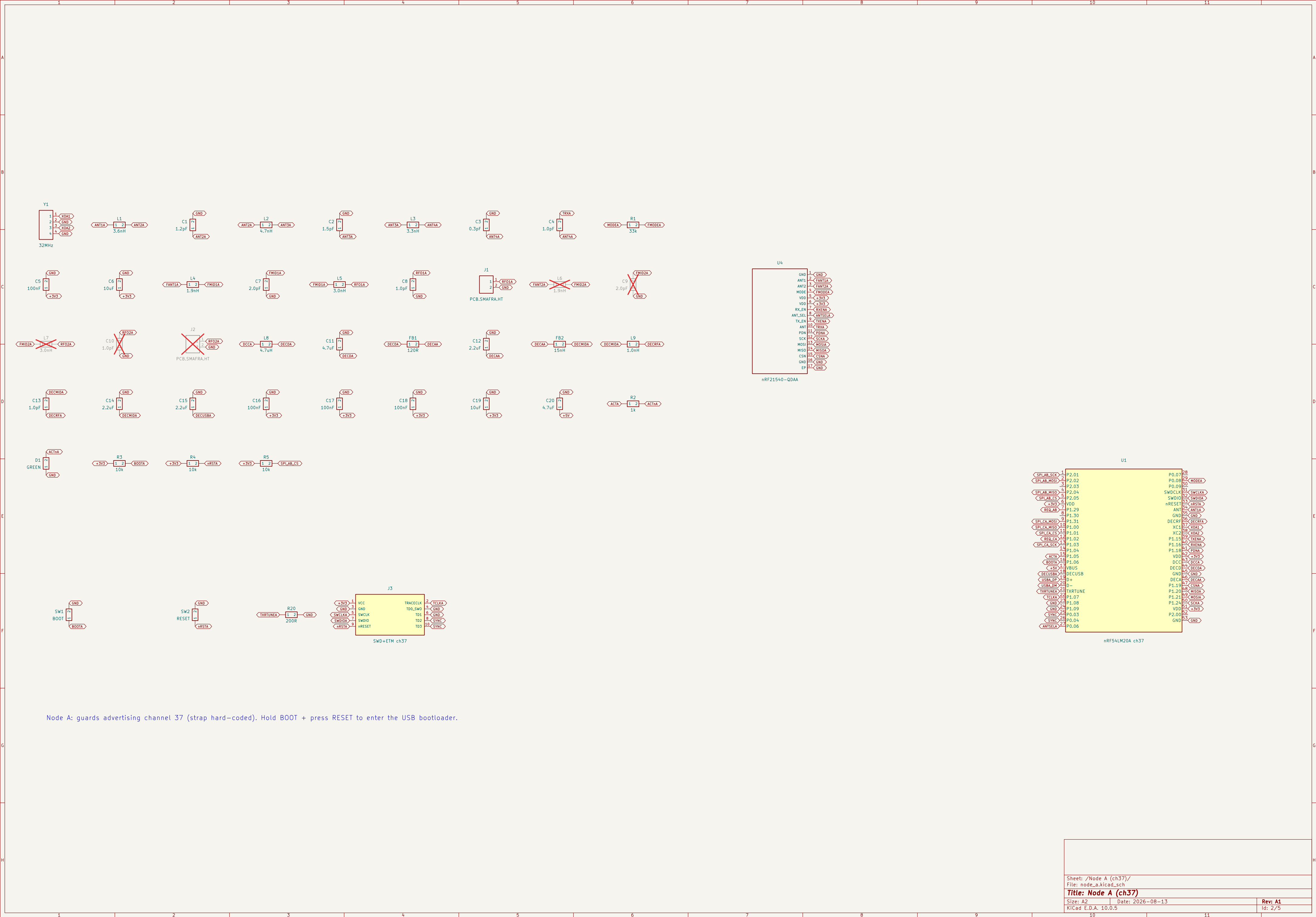
Sheet: /
File: blehound.kicad_sch

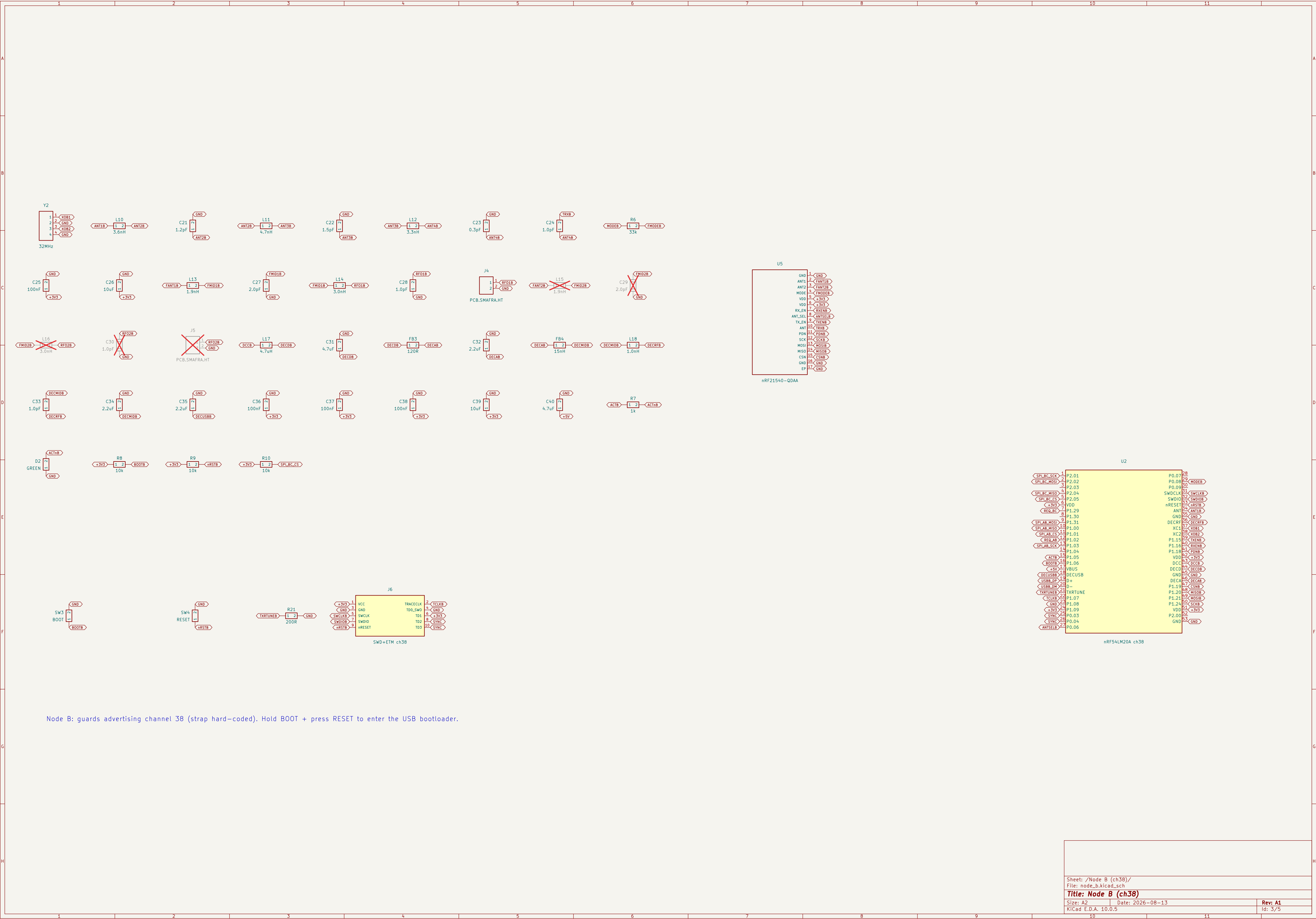
Title: blehound (root)

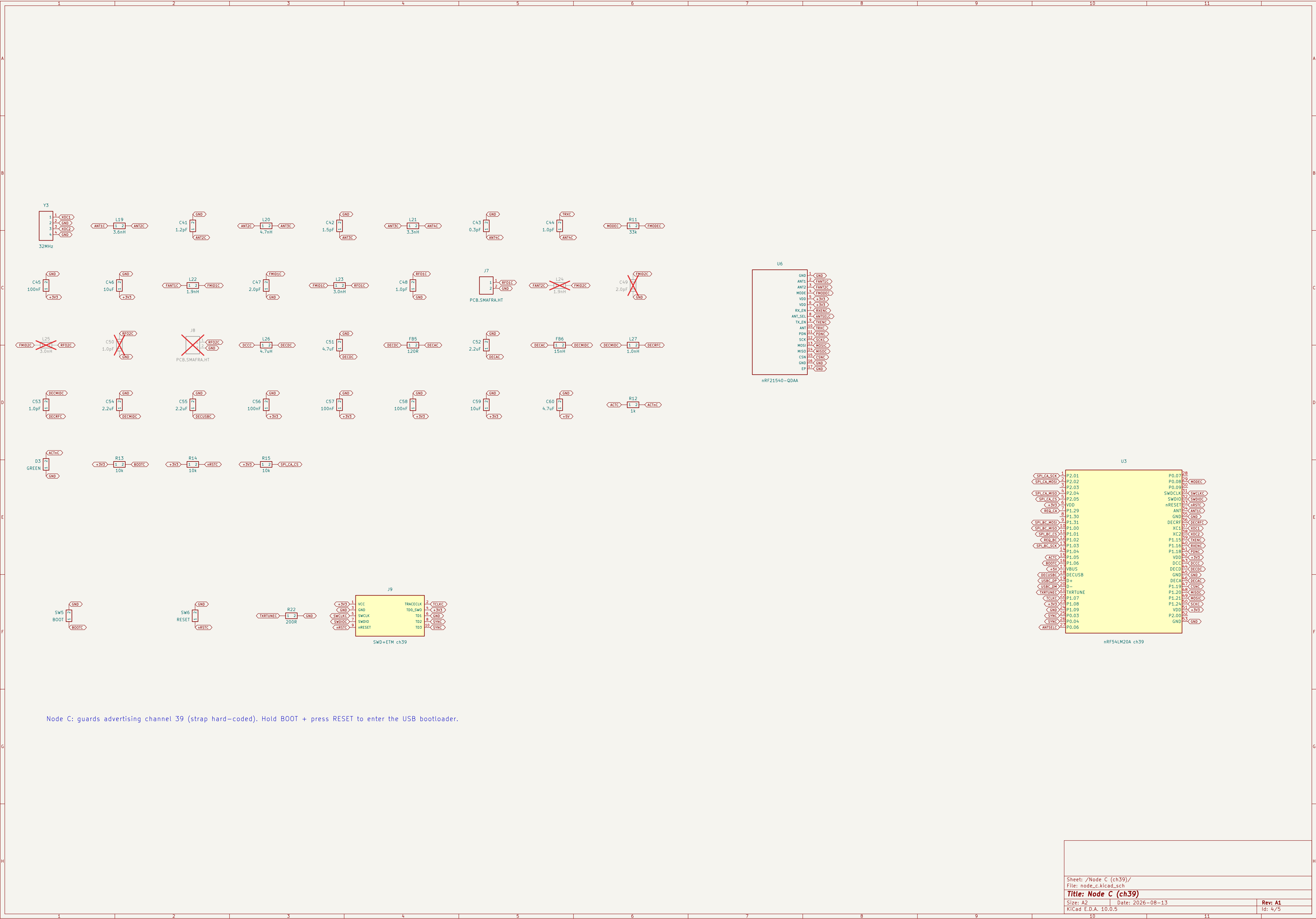
Size: A3Date: 2026-08-13

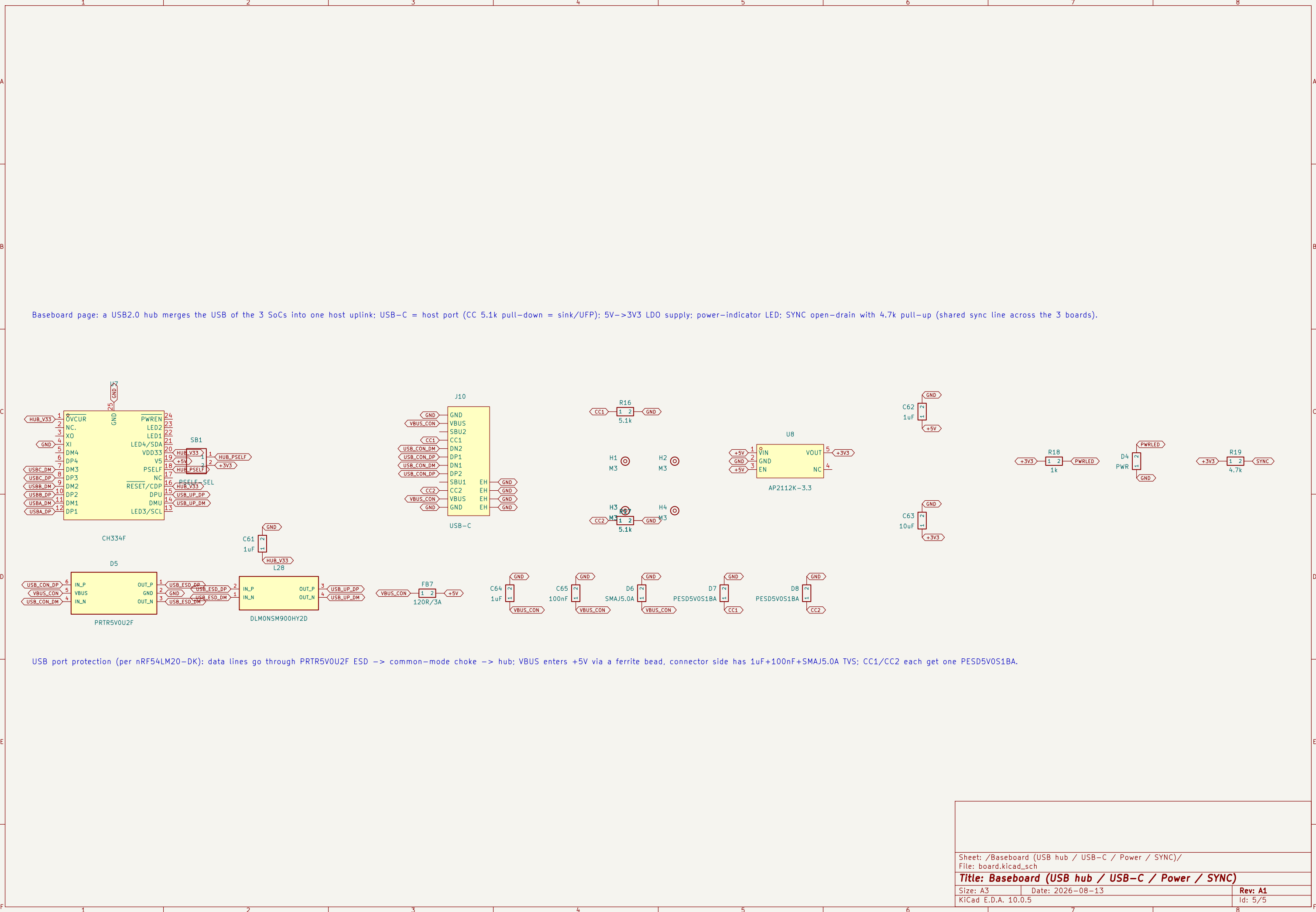
Rev: A1

KiCad E.D.A. 10.0.5Id: 1/5









Baseboard page: a USB2.0 hub merges the USB of the 3 SoCs into one host uplink; USB-C = host port (CC 5.1k pull-down = sink/UFP); 5V->3V3 LDO supply; power-indicator LED; SYNC open-drain with 4.7k pull-up (shared sync line across the 3 boards).

USB port protection (per nRF54LM20-DK): data lines go through PRTR5V0U2F ESD -> common-mode choke -> hub; VBUS enters +5V via a ferrite bead, connector side has 1uF+100nF+SMAJ5.0A TVS; CC1/CC2 each get one PESD5V0S1BA.